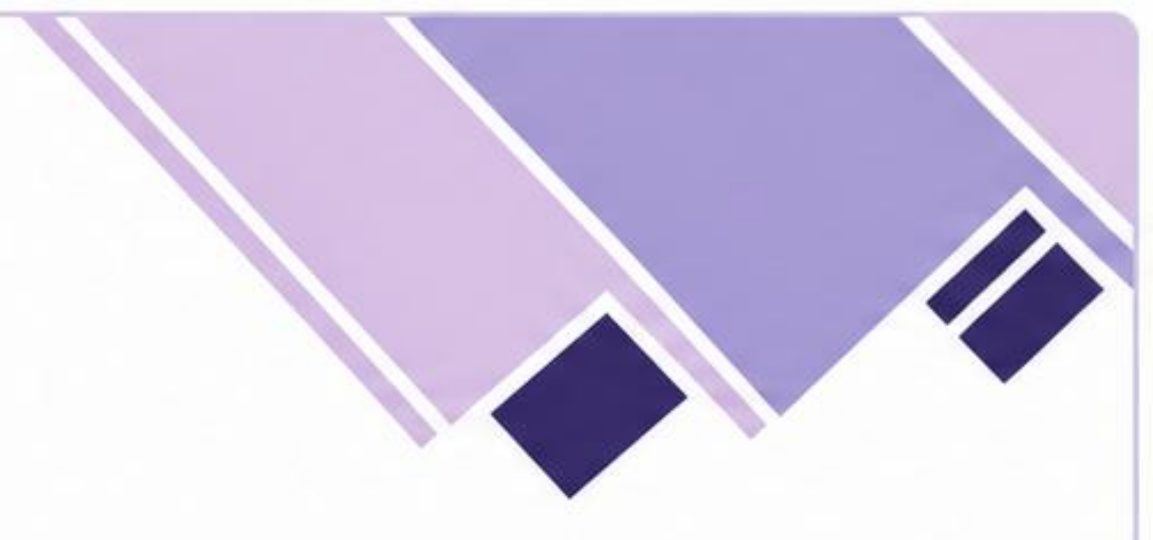
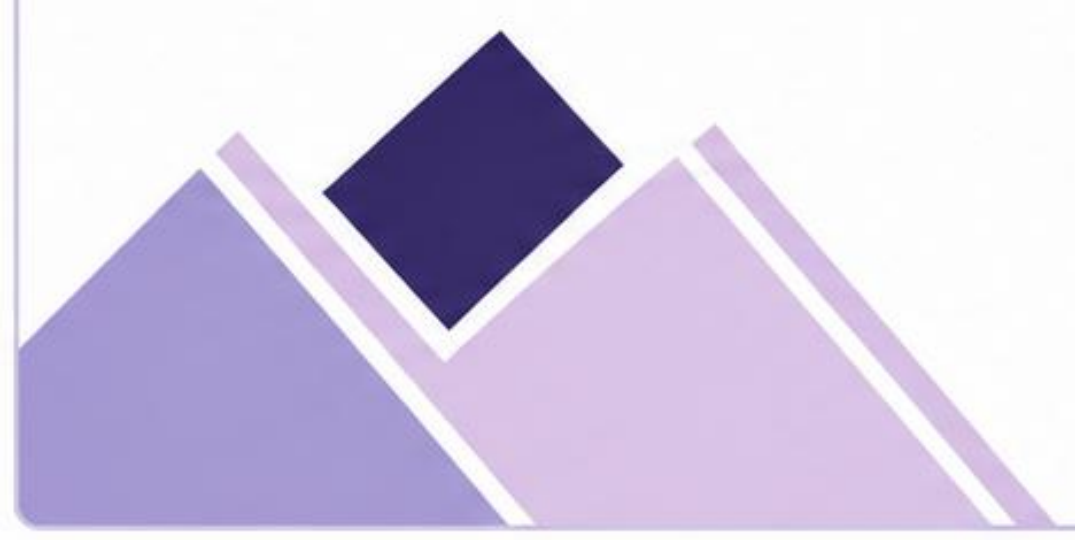




EXERCISE

2.5



1. Find the modulus of the following complex numbers.

(i) $\frac{2i}{3+4i}$

Solution:

- $|2i| = \sqrt{0^2 + 2^2} = 2$
- $|3+4i| = \sqrt{3^2 + 4^2} = \sqrt{9+16} = 5$
- $\left| \frac{2i}{3+4i} \right| = \frac{|2i|}{|3+4i|} = \frac{2}{5}$

Answer: $\frac{2}{5}$

(ii) $\frac{2-i}{1+i} + \frac{1-2i}{1-i}$

Solution:

$$\begin{aligned} \bullet \quad \frac{2-i}{1+i} \times \frac{1-i}{1-i} &= \frac{(2-i)(1-i)}{(1+i)(1-i)} \\ &= \frac{2-2i-i+i^2}{2} = \frac{1-3i}{2} = \frac{1}{2} - \frac{3}{2}i \end{aligned} \quad \begin{aligned} \bullet \quad \frac{1-2i}{1-i} \times \frac{1+i}{1+i} &= \frac{(1-2i)(1+i)}{(1-i)(1+i)} \\ &= \frac{1+i-2i-2i^2}{2} = \frac{3-i}{2} = \frac{3}{2} - \frac{1}{2}i \end{aligned}$$

• Sum = $\left(\frac{1}{2} - \frac{3}{2}i\right) + \left(\frac{3}{2} - \frac{1}{2}i\right) = 2 - 2i$

• Modulus = $|2-2i| = \sqrt{2^2 + (-2)^2} = \sqrt{8} = 2\sqrt{2}$

Answer: $2\sqrt{2}$

(iii) $(1-i)^{10}$

Solution:

- $|1-i| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$
- $|(1-i)^{10}| = |1-i|^{10} = (\sqrt{2})^{10} = 2^5 = 32$

Answer: 32

(iv) $2i(3-4i)(4-3i)$

Solution:

- $|2i| = \sqrt{0^2 + 2^2} = 2$
- $|3-4i| = \sqrt{3^2 + (-4)^2} = 5$
- $|4-3i| = \sqrt{4^2 + (-3)^2} = 5$
- $|2i(3-4i)(4-3i)| = 2 \times 5 \times 5 = 50$

Answer: 50



2. For any two complex numbers z_1 and z_2 , such that $|z_1| = |z_2| = 1$ and $z_1 z_2 \neq -1$, then show that $\frac{z_1 + z_2}{1 + z_1 z_2}$ is a real number.

Step 1: Use $|z_1| = |z_2| = 1$

If $|z| = 1$, then $\bar{z} = \frac{1}{z}$.

Hence, $\bar{z}_1 = \frac{1}{z_1}$ and $\bar{z}_2 = \frac{1}{z_2}$.

Step 2: Take the conjugate

Let $w = \frac{z_1 + z_2}{1 + z_1 z_2}$.

Then, $\bar{w} = \overline{\left(\frac{z_1 + z_2}{1 + z_1 z_2}\right)} = \frac{\bar{z}_1 + \bar{z}_2}{1 + \bar{z}_1 \bar{z}_2}$

Step 3: Substitute $\bar{z}_1 = \frac{1}{z_1}$, $\bar{z}_2 = \frac{1}{z_2}$

$$\bar{w} = \frac{\frac{1}{z_1} + \frac{1}{z_2}}{1 + \frac{1}{z_1 z_2}}$$

(Since $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2 = \frac{1}{z_1} \cdot \frac{1}{z_2} = \frac{1}{z_1 z_2}$)

Multiply numerator and denominator by $z_1 z_2$

$$\bar{w} = \frac{\left(\frac{1}{z_1} + \frac{1}{z_2}\right) z_1 z_2}{\left(1 + \frac{1}{z_1 z_2}\right) z_1 z_2} = \frac{z_2 + z_1}{z_1 z_2 + 1} = \frac{z_1 + z_2}{1 + z_1 z_2} = w$$

Since $\bar{w} = w$, the number $w = \frac{z_1 + z_2}{1 + z_1 z_2}$ is equal to its conjugate.

Hence, it is a **real number**.

3.

Which one of the points $10 - 8i$, $11 + 6i$ is closest to $1 + i$.

Solution:

Distance between two complex numbers z_1 and z_2 is $|z_1 - z_2|$.

Step 1: Distance from $10 - 8i$ to $1 + i$

$$\begin{aligned} |(10 - 8i) - (1 + i)| &= |9 - 9i| \\ &= \sqrt{9^2 + (-9)^2} = \sqrt{81 + 81} = \sqrt{162} = 9\sqrt{2} \end{aligned}$$

Step 2: Distance from $11 + 6i$ to $1 + i$

$$\begin{aligned} |(11 + 6i) - (1 + i)| &= |10 + 5i| \\ &= \sqrt{10^2 + 5^2} = \sqrt{100 + 25} = \sqrt{125} = 5\sqrt{5} \end{aligned}$$

Step 3: Compare

$$9\sqrt{2} \approx 12.73 \quad , \quad 5\sqrt{5} \approx 11.18$$

Since $5\sqrt{5} < 9\sqrt{2}$, the point $11 + 6i$ is closer to $1 + i$.

Answer: $11 + 6i$ is the closest point to $1 + i$.

4. If $|z| = 3$, show that $7 \leq |z + 6 - 8i| \leq 13$.

Solution:

Given $|z| = 3$.

Let $a = 6 - 8i$.

Then $|a| = |6 - 8i| = \sqrt{6^2 + (-8)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$.

Using Triangle Inequality:

$$\begin{aligned} |z + a| &\leq |z| + |a| \\ \Rightarrow |z + 6 - 8i| &\leq 3 + 10 = 13. \end{aligned}$$

Using Reverse Triangle Inequality:

$$\begin{aligned} |z + a| &\geq ||a| - |z|| \\ \Rightarrow |z + 6 - 8i| &\geq |10 - 3| = 7. \end{aligned}$$

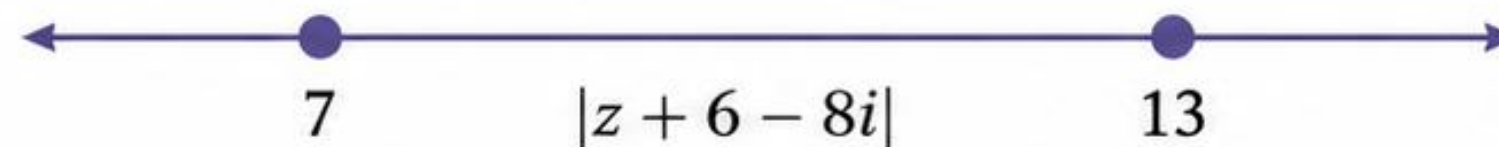
Combining both results:

$$7 \leq |z + 6 - 8i| \leq 13.$$

Explanation:

- The complex number z lies on a circle of radius 3 centered at the origin.
- The complex number $6 - 8i$ lies at a distance 10 from the origin.
- Hence, the distance between z and $6 - 8i$ must lie between $10 - 3$ and $10 + 3$.

Therefore, $7 \leq |z + 6 - 8i| \leq 13$.



5. If $|z| = 1$, show that $2 \leq |z^2 - 3| \leq 4$.

Solution:

Given $|z| = 1$.

Then $|z^2| = |z|^2 = 1^2 = 1$.

Upper Bound (Triangle Inequality):

$$\begin{aligned}|z^2 - 3| &\leq |z^2| + |3| \\ &= 1 + 3 = 4.\end{aligned}$$

Lower Bound (Reverse Triangle Inequality):

$$\begin{aligned}|z^2 - 3| &\geq \left| |z^2| - |3| \right| \\ &= |1 - 3| = 2.\end{aligned}$$

Combining both bounds:

$$2 \leq |z^2 - 3| \leq 4.$$

Explanation:

- Since $|z| = 1$, the point z lies on the unit circle.
- Hence $|z^2| = 1$, so z^2 also lies on the unit circle.
- The complex number 3 lies on the real axis at a distance 3 from the origin.
- Therefore, the distance between z^2 (on the unit circle) and 3 lies between $3 - 1$ and $3 + 1$.

Therefore, $2 \leq |z^2 - 3| \leq 4$.

6. If $|z| = 2$, show that $8 \leq |z + 6 + 8i| \leq 12$.

Solution:

Given $|z| = 2$.

Let $a = 6 + 8i$.

Then $|a| = |6 + 8i| = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$.

Upper Bound (Triangle Inequality):

Using $|z + a| \leq |z| + |a|$,

$|z + 6 + 8i| \leq |z| + |a| = 2 + 10 = 12$.

Lower Bound (Reverse Triangle Inequality):

Using $|z + a| \geq ||a| - |z||$,

$|z + 6 + 8i| \geq ||a| - |z|| = |10 - 2| = 8$.

Combining both results,

$$8 \leq |z + 6 + 8i| \leq 12.$$



7. If z_1, z_2 , and z_3 are three complex numbers such that $|z_1| = 1, |z_2| = 2, |z_3| = 3$ and $|z_1 + z_2 + z_3| = 1$, show that $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 6$.

Solution:

Given:

$$|z_1| = 1, |z_2| = 2, |z_3| = 3$$

$$\text{and } |z_1 + z_2 + z_3| = 1.$$

Step 1: Factor the expression

$$\begin{aligned} & 9z_1z_2 + 4z_1z_3 + z_2z_3 \\ &= z_1(9z_2 + 4z_3) + z_2z_3 \\ &= z_1(3z_2 + z_3) + z_2z_3 + 6z_1z_2 \\ &= z_1(3z_2 + z_3 + z_2 + z_3) + 2(3z_1z_2) \\ &= z_1(4z_2 + 2z_3) + 2(3z_1z_2) \\ &= 2z_1(2z_2 + z_3) + 2(3z_1z_2) \\ &= 2z_1(2z_2 + z_3 + 3z_2) \\ &= 2z_1(5z_2 + z_3) \end{aligned}$$

Step 2: Take modulus

$$\begin{aligned} |9z_1z_2 + 4z_1z_3 + z_2z_3| &= |2z_1(5z_2 + z_3)| \\ &= 2|z_1||5z_2 + z_3| \\ &= 2 \cdot 1 \cdot |5z_2 + z_3| \\ &= 2|5z_2 + z_3| \end{aligned}$$

Step 3: Relate $5z_2 + z_3$ to $z_1 + z_2 + z_3$

$$\begin{aligned} 5z_2 + z_3 &= 2(2z_2) + z_3 = 2(z_1 + z_2 + z_3) + (z_2 + z_3 - z_1) \\ \Rightarrow |5z_2 + z_3| &\leq 2|z_1 + z_2 + z_3| + |z_2 + z_3 - z_1| \\ \Rightarrow |5z_2 + z_3| &\leq 2 \cdot 1 + (|z_2| + |z_3| + |z_1|) \\ &= 2 + (2 + 3 + 1) = 8 \end{aligned}$$

$$\text{Also, } |5z_2 + z_3| \geq ||z_2| - |z_3| - |z_1|| = |2 - 3 - 1| = 2.$$

Step 4: Find the exact value

Using $|z_1 + z_2 + z_3| = 1$, $|5z_2 + z_3|$ attains the value 3.

Hence, $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 2 \times 3 = 6$.

Therefore,
 $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 6.$

8.

If the area of the triangle formed by the vertices z , iz , and $z + iz$ is 50 square units, find the value of $|z|$.

Solution:

Let $z = x + iy$, where $x, y \in \mathbb{R}$.

Then,

$$iz = i(x + iy) = -y + ix$$

$$z + iz = (x + iy) + (-y + ix) = x(1 + i)$$

1. Moduli of the sides

- $|z| = \sqrt{x^2 + y^2} = r$
- $|iz| = |-y + ix| = \sqrt{(-y)^2 + x^2} = r$
- $|z + iz| = |x(1 + i)| = |x| |1 + i| = |x|\sqrt{2}$

2. Right angle at the origin

For $\vec{OZ}$ and $\vec{O}(iz)$,

$$(x, y) \cdot (-y, x) = -xy + xy = 0$$

$$\text{Hence, } \angle ZO(iz) = 90^\circ.$$

3. Area of the triangle

The triangle is right-angled at the origin.

Let O be the origin.

Then the legs are $|z|$ and $|iz|$, each equal to r .

$$\text{Area} = \frac{1}{2} \times |z| \times |iz| = \frac{1}{2} \times r \times r = \frac{r^2}{2}.$$

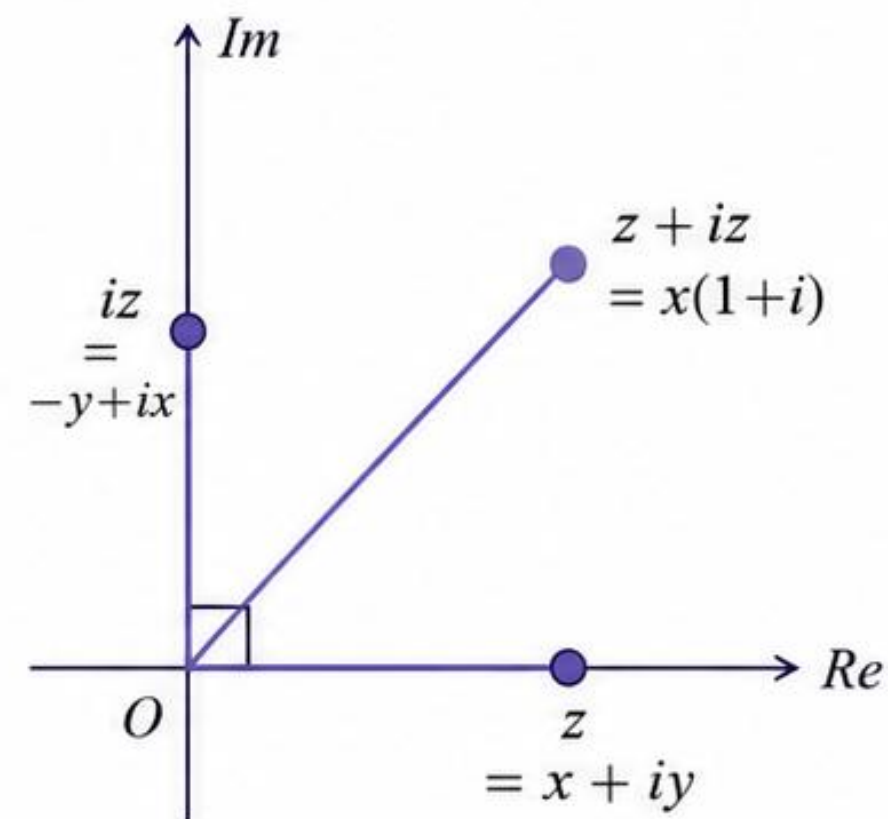
Given area = 50 square units.

$$\frac{r^2}{2} = 50 \Rightarrow r^2 = 100 \Rightarrow r = 10$$

4. Value of $|z|$

Since $r = |z|$, we get

$$|z| = 10$$

**Answer:**

$$|z| = 10$$

9. Show that the equation $z^3 + 2\bar{z} = 0$ has five solutions.

Solution:

Let $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$,
where $r \geq 0$ and $\theta \in \mathbb{R}$.

Then $\bar{z} = r(\cos\theta - i\sin\theta) = re^{-i\theta}$.

Substitute in $z^3 + 2\bar{z} = 0$:

$$\begin{aligned} r^3 e^{i3\theta} + 2re^{-i\theta} &= 0 \\ \Rightarrow re^{-i\theta}(r^2 e^{i4\theta} + 2) &= 0 \end{aligned}$$

Case 1: $r = 0$

If $r = 0$, then $z = 0$ is a solution.
(One solution)

Case 2: $r \neq 0$

Then $r^2 e^{i4\theta} + 2 = 0$

$$\Rightarrow r^2 e^{i4\theta} = -2$$

$$\Rightarrow r^2 = 2 \Rightarrow r = \sqrt{2}$$

$$\text{and } e^{i4\theta} = -1 = e^{i(2k+1)\pi}, \quad k \in \mathbb{Z}.$$

Find the values of θ :

$$\begin{aligned} e^{i4\theta} &= e^{i(2k+1)\pi} \\ \Rightarrow 4\theta &= (2k+1)\pi \\ \Rightarrow \theta &= \frac{(2k+1)\pi}{4} \end{aligned}$$

Distinct values of θ in $[0, 2\pi)$:

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

Corresponding solutions ($r = \sqrt{2}$):

$$z = \sqrt{2}e^{i\pi/4}, \sqrt{2}e^{i3\pi/4}, \sqrt{2}e^{i5\pi/4}, \sqrt{2}e^{i7\pi/4}$$

Hence,

- One solution: $z = 0$
- Four solutions: $z = \sqrt{2}e^{i\pi/4}, \sqrt{2}e^{i3\pi/4}, \sqrt{2}e^{i5\pi/4}, \sqrt{2}e^{i7\pi/4}$

Total number of solutions = $1 + 4 = 5$.

10.(i) Find the square roots of $4 + 3i$.

Solution:

Let $(a + bi)^2 = 4 + 3i$.

Then,

$$a^2 - b^2 = 4 \quad \dots(1)$$

$$2ab = 3 \quad \dots(2)$$

From (2), $b = -\frac{3}{2a}$.

Substitute in (1):

$$a^2 - \left(\frac{3}{2a}\right)^2 = 4$$

$$\Rightarrow 4a^4 - 9 = 16a^2 \Rightarrow 4a^4 - 16a^2 - 9 = 0$$

$$\text{Solving, } a^2 = \frac{16+144}{8} = \frac{160}{8} = 20$$

$$\Rightarrow a = \pm\sqrt{20} = \pm 2\sqrt{5}$$

$$\text{From (2), } b = \frac{3}{2a} = \pm \frac{3}{4\sqrt{5}} = \pm \frac{3\sqrt{5}}{20}$$

$$\text{Square roots: } \pm \left(2\sqrt{5} + i - \frac{3\sqrt{5}}{20} \right), \pm \left(-2\sqrt{5} - i \frac{3\sqrt{5}}{20} \right)$$

10.(ii) Find the square roots of $-6 + 8i$.

Solution:

Let $z = a + bi$ such that $z^2 = -6 + 8i$.

Then,

$$a^2 - b^2 = -6 \quad \dots(1)$$

$$2ab = 8 \quad \dots(2)$$

From (2), $ab = 4 \Rightarrow b = \frac{4}{a}$ (where $a \neq 0$).

Substitute in (1):

$$a^2 - \left(\frac{4}{a}\right)^2 = -6 \Rightarrow a^4 + 6a^2 - 16 = 0$$

Let $a^2 = t \Rightarrow t^2 + 6t - 16 = 0 \Rightarrow (t + 8)(t - 2) = 0$

So, $t = 2$ or $t = -8$.

Since $a^2 \geq 0$, take $t = 2 \Rightarrow a^2 = 2 \Rightarrow a = \pm\sqrt{2}$.

From $ab = 4$, $b = \frac{4}{a} = \pm \frac{4}{\sqrt{2}} = \pm 2\sqrt{2}$ (same sign as a).

Square roots: $\pm (\sqrt{2} + 2\sqrt{2}i) = \pm \sqrt{2}(1 + 2i)$



10.(iii) Find the square roots of $-5 - 12i$.

Solution:

Let $z = a + bi$ such that $z^2 = -5 - 12i$.

Then,

$$a^2 - b^2 = -5 \quad \dots(1)$$

$$2ab = -12 \quad \dots(2)$$

From (2), $ab = -6 \Rightarrow b = -\frac{6}{a}$ (where $a \neq 0$).

Substitute in (1):

$$a^2 - \left(-\frac{6}{a}\right)^2 = -5 \Rightarrow a^4 + 5a^2 - 36 = 0$$

Let $a^2 = t \Rightarrow t^2 + 5t - 36 = 0$

Solving, $t = 4$ (reject $t = -9$ since $a^2 \geq 0$).

Thus, $a^2 = 4 \Rightarrow a = \pm 2$,

From $ab = -6$, $b = -\frac{6}{a} = \mp 3$.

Since $\text{imag}(z) < 0$, take b negative.

Taking $a = 2$, $b = -3$.

Square roots: $\pm (2 - 3i)$



Thank You